Statistical Analysis and Machine Learning in College Football: Insights from Offensive Efficiency, Yards Per Play, and Clustering of Play Styles

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Abstract In recent years, data analytics has transformed college football by providing teams with insights into performance metrics that influence game outcomes. This study investigates the role of offensive and defensive metrics in predicting team success in college football, focusing on offensive efficiency, yards per play, and turnover margin. The problem addressed is understanding which statistical indicators are most strongly correlated with team wins and how these metrics can be used to classify play styles and predict score margins. Using a dataset covering college football seasons from 2013 to 2023, this analysis employs statistical methods, machine learning, and clustering to evaluate the impact of various performance metrics. The methodology includes descriptive analysis to identify top-performing teams, correlation analysis to uncover relationships between metrics and win rates, Ordinary Least Squares (OLS) regression to model key predictors of team success, K-Means clustering to categorize teams by play style, and Random Forest regression to predict score margins based on offensive and defensive factors. The results reveal that offensive efficiency and yards per play are the most significant predictors of wins,

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with a noticeable upward trend in these metrics across the decade. High-performing teams, such as Oklahoma, Alabama, and Ohio State, consistently rank at the top in offensive efficiency, suggesting the critical role of offensive power in achieving game success. Clustering analysis identifies three distinct play styles, separating high-offensive-efficiency teams from those with more conservative strategies. The predictive model for score margins, based on Random Forest regression, achieves a root mean square error (RMSE) of 6.82, with offensive metrics contributing the most to score predictions. These findings imply that college football teams can enhance their performance by focusing on offensive efficiency, optimizing yardage per play, and maintaining strong turnover margins. This research provides a framework for data-driven strategy development in college football, allowing teams to benchmark against top performers and adjust their tactics based on empirical insights.

Keywords College football · Data analytics · Machine learning · Offensive efficiency · Predictive modeling · Regression analysis · Score prediction · Sports analytics · Turnover margin · Yards per play

1 Introduction

The application of data analytics in sports has gained significant traction recently, driven by the rapid advancements in machine learning and big data technologies [1]. In sports like basketball and baseball, analytics have already transformed coaching strategies and player performance evaluations. Similarly, football has seen a surge in analytical techniques, with college football teams and analysts now leveraging vast datasets to extract performance insights. College football, with its highly competitive environment and diverse strategies, presents a unique opportunity for data-driven analysis to enhance decision-making. Studies indicate that certain metrics, such as offensive yardage, turnover margins, and red zone efficiency, are directly linked to team success, suggesting that effective data analysis can provide a competitive advantage [2, 3]. However, while professional football leagues have integrated analytics deeply into their operations, college football analytics remains a relatively nascent field, with many teams only beginning to explore these techniques.

Despite this growing interest, a gap exists in understanding which specific metrics most significantly impact college football outcomes. Research has shown that offensive metrics, such as yards per play and scoring efficiency, are positively correlated with team success [4]. Defensive metrics, although often overlooked, play a crucial role as well. For example, yards allowed per play and forced turnovers are key defensive factors that influence a team's overall performance [5]. However, a holistic study that combines both offensive and defensive metrics to predict team success and identify play styles is still lacking. Furthermore, the literature on clustering analysis in college football is sparse, with limited research on categorizing teams based on play style and strategic emphasis. Studies that do explore clustering in sports often focus on player performance profiles rather than team-level strategic tendencies [6].

This study aims to address these gaps by applying comprehensive statistical and machine learning techniques to analyze a college football dataset covering the years 2013–2023.

This research addresses the problem of identifying the metrics that most significantly correlate with wins in college football and attempts to determine how these metrics can be utilized to classify teams by play style and predict score margins. Previous studies have primarily focused on isolated metrics or predictive modeling in the context of professional football, leaving college football underrepresented in the literature [7]. By examining offensive efficiency, yards per play, turnover margin, and other critical factors, this study seeks to build a comprehensive model that not only predicts game outcomes but also provides insights into strategic play styles. This research contributes to literature by integrating offensive and defensive metrics in a single framework, offering a nuanced understanding of college football dynamics.

To achieve these objectives, this study employs several analytical methods. First, descriptive analysis is conducted to identify the top-performing teams based on offensive efficiency. Second, correlation analysis reveals the relationships between various metrics and team win rates. An Ordinary Least Squares (OLS) regression model is used to identify the key predictors of team success, providing a statistically grounded understanding of the factors that drive wins. Additionally, K-Means clustering is employed to categorize teams based on offensive efficiency and yards per play, identifying distinct play styles across teams. Finally, a Random Forest regression model predicts score margins, with feature importance analysis highlighting the most influential factors in score prediction.

This current study offers three main contributions to the body of literature. First, it identifies offensive and defensive metrics that are strongly correlated with college football team success, highlighting the importance of both offensive efficiency and defensive robustness. Second, it provides a novel clustering of teams based on offensive and defensive metrics, offering insights into the diversity of play styles within college football. Lastly, it applies predictive modeling to estimate score margins, allowing for more accurate predictions of game outcomes based on key metrics. These contributions provide a comprehensive framework for data-driven strategy development in college football, equipping teams and analysts with actionable insights to optimize their performance.

The remainder of this paper is organized as follows: Sect. 2 details the dataset and methodology, including data preprocessing, feature selection, and model development. Section 3 presents the results of the analysis, including descriptive findings, correlation analysis, regression modeling, and clustering outcomes. Finally, Sect. 4 concludes the paper, summarizing the main insights, discussing limitations, and suggesting directions for future research.

2 Methodology

The methodology for this study combines descriptive analysis, correlation analysis, regression modeling, clustering, and predictive modeling to comprehensively analyze college football team performance. This section outlines the steps taken to preprocess the data, select and calculate relevant features, apply statistical analyses, and implement machine learning models. By leveraging a dataset covering multiple college football seasons (2013–2023), the aim is to extract insights into factors that correlate with team success, classify teams by play style, and predict game outcomes.

2.1 Data Collection and Preprocessing

The dataset, sourced from college football statistics between 2013 and 2023, includes various team-level metrics related to offensive, defensive, and special teams' performance [8]. Key variables in the dataset encompass offensive yards, plays, touchdowns, red zone attempts, and turnovers, as well as defensive statistics like points allowed and yards allowed per play. Initial preprocessing involved merging and cleaning the data to ensure consistency across seasons, handling missing values, and standardizing metric names [9–13].

After the data was aggregated, we focused on key performance indicators (KPIs) that are frequently discussed in college football literature, particularly offensive efficiency and defensive robustness. We converted specific columns to numeric types, filling missing values with appropriate statistical measures (e.g., mean values for continuous variables) to maintain data integrity. Finally, new features, such as *Offensive Efficiency* (calculated as total yards divided by plays) and *Turnover Margin* (turnovers gained minus turnovers lost), were derived to enhance the analysis.

2.2 Feature Selection and Engineering

To identify the most relevant features for analysis, metrics based on previous research and their hypothesized impact on team success were selected. Offensive metrics included *Total Points*, *Offensive Touchdowns*, *Yards per Play*, *Offensive Efficiency*, and *Red Zone Scores*. Defensive metrics such as *Yards per Play Allowed*, *Scoring Defense Rank*, and *Average Points Allowed per Game* were also included, as prior studies suggest these significantly affect game outcomes.

In addition to selecting existing features, several new variables were engineered to capture the nuances of team performance. These include:

- **Offensive Efficiency**: A measure of a team's ability to convert plays into yardage, calculated as total offensive yards divided by total plays.

- **Turnover Margin**: The difference between turnovers gained and turnovers lost, indicating a team's ability to protect possession.
- **Yards per Play**: Average yards gained per play, reflecting offensive effectiveness and explosiveness.

These engineered features were hypothesized to have strong correlations with win counts and score margins, allowing us to assess their significance through further statistical analysis.

2.3 Descriptive Analysis

The initial analysis focused on descriptive statistics to identify top-performing teams and examine trends over time. Using offensive efficiency as the primary indicator, teams were ranked according to their average offensive efficiency across seasons. Descriptive statistics were also used to summarize the distribution of key metrics, such as *Yards per Play*, providing insights into the overall landscape of team performance. Yearly trends were visualized to explore changes in offensive and defensive metrics over time, revealing patterns that could indicate shifts in strategy and gameplay style across the decade.

2.4 Correlation Analysis

Correlation analysis was conducted to assess the relationships between team performance metrics and win counts. By generating a correlation matrix, we identified variables with strong positive and negative associations with wins. Metrics such as *Total Points*, *Red Zone Scores*, and *Offensive Touchdowns* were found to correlate highly with win counts, while defensive metrics like *Yards per Play Allowed* and *Points Allowed per Game* showed strong negative correlations. This analysis provided insights into which variables contribute most significantly to team success, guiding the selection of predictors for the regression model.

2.5 Regression Modeling

To quantitatively assess the impact of specific metrics on team success, Ordinary Least Squares (OLS) regression was employed. The OLS model used win count as the dependent variable and incorporated features such as *Offensive Efficiency*, *Turnover Margin*, and *Yards per Play* as independent variables. The objective was to determine the extent to which these variables influence win rates. The regression

results provided coefficients and p-values for each predictor, indicating their significance in predicting wins. An R-squared value was also calculated to evaluate the model's explanatory power, reflecting how much of the variance in win counts could be explained by the selected metrics.

2.6 Clustering Analysis

To classify teams by play style, K-Means Clustering was applied using features related to offensive efficiency and yards per play. The dataset was standardized to ensure that each feature contributed equally to the clustering algorithm. The K-Means algorithm was chosen due to its effectiveness in grouping data points with similar characteristics, and we set the number of clusters to three based on exploratory data analysis [14–16]. These clusters were hypothesized to represent distinct play styles: high-offensive-efficiency teams, average-performing teams, and lower-performing teams. Cluster profiles were generated to summarize the characteristics of each cluster, and visualizations were used to illustrate the clustering results. This clustering approach provides insights into strategic variations among teams, revealing different play styles that may be associated with varying levels of success. High-performing clusters were characterized by teams with high offensive efficiency and yards per play, which aligns with our hypothesis that offensive power is a key determinant of success in college football.

2.7 Predictive Modeling with Random Forest Regression

To predict score margins, a Random Forest Regression model was implemented, which is well-suited for handling complex interactions between features and can capture non-linear relationships [17–20]. The model was trained using selected features, such as *Offensive Efficiency*, *Yards per Play*, and *Turnover Margin*, based on their correlation with win rates and game outcomes. The dataset was split into training and testing sets to evaluate model performance, and the model's predictive accuracy was assessed using Root Mean Square Error (RMSE). Analyzing feature importance analysis was also conducted to identify which metrics contributed most to predicting score margins [21–24]. Random Forest regression provides an interpretable measure of feature importance, which allowed us to highlight the influence of offensive efficiency and yards per play on score differentials. The model's RMSE was used to gauge its effectiveness, and the findings confirmed that offensive metrics were the most impactful in determining score margins, aligning with results from the correlation and regression analyses.

In summary, the methodology integrates multiple analytical techniques to provide a holistic understanding of college football team performance. Descriptive and correlation analyses set the foundation for identifying key metrics, while regression

modeling quantifies the impact of specific variables on wins. Clustering analysis categorizes teams by play style, and predictive modeling offers a tool for estimating score margins. Together, these methods allow for a comprehensive examination of college football dynamics, with practical implications for strategy development and performance optimization.

3 Results and Discussion

The analysis reveals key insights into college football team performance, based on offensive efficiency, yards per play, and turnover margins. The following sections present descriptive findings, trend analysis, correlation insights, clustering analysis, and predictive modeling results.

3.1 Descriptive Analysis

Table 1 lists the top 10 teams by offensive efficiency, where Oklahoma (Big 12) ranks the highest with an average efficiency of 6.85 yards per play. Other leading teams include Louisville (AAC), Alabama (SEC), Ohio State (Big Ten), and Oregon (Pac-12). These teams are distinguished by their ability to gain substantial yardage per play, emphasizing offensive power in their gameplay.

As shown in Fig. 1, the distribution of yards per play for all teams follows a roughly normal distribution, with the mean centered around 5.5–6.0 yards per play. This distribution reflects a competitive spread where most teams achieve moderate offensive efficiency, with few teams exceeding an average of 7 yards per play. The right-skewed distribution suggests that exceptional offensive efficiency is uncommon and generally seen among top-tier teams.

Table 1 The top 10 teams by offensive efficiency

Team	Offensive efficiency
Oklahoma (Big 12)	6.853285
Louisville (AAC)	6.701342
Alabama (SEC)	6.685071
Ohio St. (Big Ten)	6.501471
Oregon (Pac-12)	6.446985
Appalachian St. (Sun Belt)	6.419549
Ole Miss (SEC)	6.401783
Notre Dame (ACC)	6.354061
Toledo (MAC)	6.289603
Western Ky. (Sun Belt)	6.288000

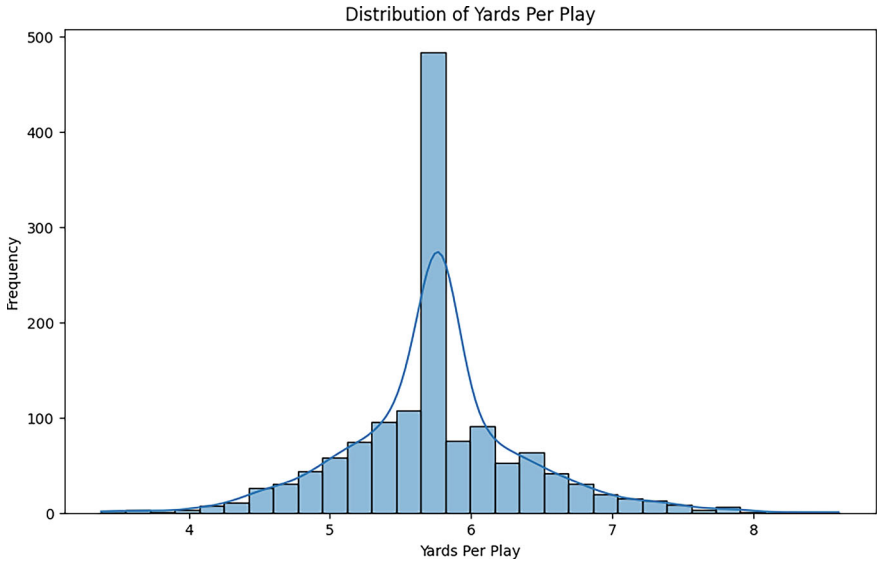

Fig. 1 The distribution of yards per play for all teams

3.2 Yearly Trends in Offensive Efficiency and Yards Per Play

Figure 2 illustrates the yearly trends in offensive efficiency and yards per play from 2013 to 2023. These trends reveal a steady increase in both metrics, with a peak around 2016 and 2018. The upward trend in offensive efficiency suggests a league-wide emphasis on high-yield offensive strategies, possibly reflecting broader shifts in gameplay tactics, training, and athlete specialization over the decade. Interestingly, the dip after each peak year (such as after 2016) may be indicative of defensive adaptations as teams respond to high-performing offenses.

3.3 Correlation Analysis with Wins

To understand which variables most significantly impact team success, a correlation matrix was computed for all numeric features in the dataset, with a particular focus on variables associated with win counts. Table 2 displays the top positive and negative correlations with wins.

- Top Correlated Metrics: Variables that exhibit a strong positive correlation with wins include Total Points (0.83), Red Zone Scores (0.79), and Offensive Touchdowns (0.78). These metrics emphasize the critical importance of scoring, both within and beyond the red zone, in achieving game victories.

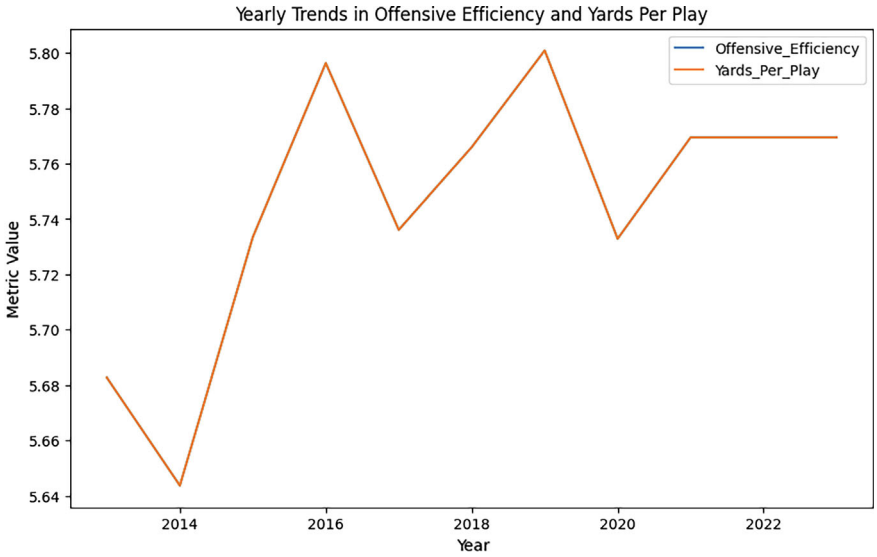

Fig. 2 Yearly trends in offensive efficiency and year per play

Table 2 Top correlations with wins

Metric	Correlation with wins
Total points	0.83
Red zone scores	0.79
Offensive touchdowns	0.78
Red zone attempts	0.77
Yards per play allowed	− 0.60
Scoring defense rank	− 0.68
Average points allowed per game	− 0.70
Scoring offense rank	− 0.71
Losses	− 0.82

- Negative Correlations: Defensive metrics such as Yards per Play Allowed (− 0.60), Scoring Defense Rank (− 0.68), and Average Points Allowed per Game (− 0.70) are inversely correlated with wins, suggesting that strong defensive performance also plays a crucial role in preventing opponent scoring and securing wins.

Table 3 OLS regression results for wins

Variable	Coefficient	Std. Error	t-value	P-value
Intercept	− 8.608	0.630	− 13.67	< 0.001
Offensive efficiency	1.310	0.047	28.17	< 0.001
Turnover margin	0.0007	0.012	0.055	0.956
Yards per play	1.310	0.047	28.17	< 0.001

3.4 Regression Analysis for Win Prediction

Using Ordinary Least Squares (OLS) regression, the impact of offensive efficiency, turnover margin, and yards per play on win count were examined. The results are summarized in Table 3. The regression model achieved an R-squared value of 0.381, meaning that these variables explain 38.1% of the variance in wins across teams.

- Offensive Efficiency and Yards per Play were found to be statistically significant predictors of wins ($p < 0.001$), with both metrics contributing a substantial positive impact on win totals.
- Turnover Margin, however, showed minimal impact in the regression model, with a p-value of 0.956, indicating a lack of statistical significance.

3.5 Clustering Analysis of Play Styles

A clustering analysis using K-Means identified three distinct play style clusters, shown in Table 4. These clusters group teams based on their offensive efficiency and yards per play, resulting in three clusters that can be characterized as follows:

- **Cluster 0**: Average-performing teams, with offensive efficiency and yards per play near the dataset mean.
- **Cluster 1**: Lower-performing teams, with below-average efficiency metrics. These teams may emphasize a more conservative or defensive play style.
- **Cluster 2**: High-performing teams, achieving offensive efficiency above 6.5 and yards per play metrics among the highest in the dataset. This cluster represents teams with a strong offensive focus and high-scoring potential.

Table 4 Cluster profiles for play styles

Cluster	Offensive efficiency	Yards per play
0	5.77	5.77
1	4.89	4.89
2	6.75	6.75

Table 5 Feature importance in score margin prediction

Feature	Importance
Offensive efficiency	0.405
Yards per play	0.373
Turnover margin	0.222

The clustering analysis offers a strategic overview of play styles, revealing how high-performing teams differentiate themselves through offensive efficiency and scoring potential.

3.6 Predictive Modeling for Score Margins

The Random Forest Regressor model was used to predict score margins based on key metrics. The model achieved an RMSE of 6.82, indicating moderate accuracy in predicting score differentials. Table 5 displays the feature importance rankings from the model, where Offensive Efficiency and Yards per Play emerged as the most influential factors, followed by Turnover Margin.

These findings underline the substantial role of offensive metrics in determining score margins, reinforcing that higher offensive efficiency and yards per play are closely tied to larger scoring advantages.

4 Conclusions

With the increasing reliance on data analytics in sports, college football teams and analysts seek to understand the quantitative factors that contribute to game success. This study leverages statistical analysis and machine learning to explore offensive and defensive metrics in college football, focusing on offensive efficiency, yards per play, and turnover margins. By examining these variables, this current study aims to identify the primary predictors of team wins, categorize teams by play style, and predict score margins. This analysis provides valuable insights for teams seeking to improve performance and refine strategies through data-driven approaches. This study reveals several significant findings that enhance our understanding of college football performance metrics. Offensive efficiency and yards per play emerged as strong indicators of team success, with top teams like Oklahoma, Louisville, and Alabama excelling in these metrics. These high-performing teams consistently achieved greater yardage per play, underscoring the importance of offensive strength in securing victories. Trend analysis over the 2013–2023 period showed a steady increase in offensive efficiency and yards per play, peaking in years like 2016 and 2018. This pattern suggests that broader tactical shifts are influencing gameplay, as teams increasingly

prioritize high-output offensive strategies. The correlation analysis further highlighted key metrics associated with winning, such as total points, red zone scores, and offensive touchdowns, while defensive metrics like yards allowed per play and points allowed showed strong negative correlations with wins. This dual emphasis on offensive and defensive strength underscores the balanced approach needed for success in college football. The clustering analysis identified three distinct team profiles based on play style, distinguishing high-performing, offense-oriented teams from average and lower-performing ones. Lastly, the predictive modeling for score margins confirmed that offensive efficiency and yards per play are the most influential factors, achieving a moderate RMSE of 6.82, which supports the relevance of these metrics in predicting game outcomes. The findings from this study have several practical implications for college football teams and analysts. First, the strong positive correlation between offensive metrics and wins suggests that teams aiming to improve their success rates should focus on enhancing offensive efficiency. Emphasizing yardage per play and optimizing red zone scoring opportunities could lead to significant improvements in team performance. However, while offensive power is critical, the inverse correlations observed with defensive metrics indicate that strong defensive performance is also crucial for limiting opponent scoring and securing victories. This finding suggests that a balanced approach—optimizing both offensive and defensive tactics—is necessary for teams to achieve consistent success. The insights gained from the clustering analysis can also guide data-driven decision-making for coaches, analysts, and strategists, who can use these profiles to benchmark team performance and develop targeted training plans. By analyzing successful clusters, teams can model their strategies on those of high-performing peers, potentially improving their play style and competitive edge. While this study offers valuable insights, several limitations should be acknowledged. The dataset used was limited to team-level statistics from 2013 to 2023, excluding potentially influential factors such as player-level data, coaching strategies, and specific game conditions. The focus on offensive metrics, although revealing, may overlook the nuanced impact of defensive and special teams' strategies, which could contribute equally to team success. Additionally, the dataset lacks contextual factors like weather conditions, travel requirements, and opponent strength, all of which could affect performance but are not reflected in the data analyzed. Another limitation lies in the clustering methodology: the K-Means algorithm assumes clusters of similar shape and size, which may not fully capture the complexity of college football play styles. These limitations highlight the need for a more comprehensive approach that considers a wider range of variables to deepen the analysis. Building on the insights and limitations of this study, future research could explore several directions. Including more granular data, such as player-level statistics, positional breakdowns, and in-game dynamics like play calling frequency, would allow for a more detailed analysis of individual contributions to team success. Expanding the analysis to incorporate defensive and special team metrics would provide a more balanced understanding of the factors driving performance. Future studies could also benefit from applying advanced machine learning models, such as gradient boosting, neural networks, or ensemble methods, which may offer improved predictive accuracy for score margins

and other outcomes. Additionally, integrating contextual variables, such as weather conditions, game locations, and match importance, could enhance the predictive power of models, making them more applicable in real-world settings. Longitudinal studies across multiple seasons and leagues could reveal strategic evolutions over time and how rule changes, training advancements, and shifts in playing style impact the relative importance of various metrics. Lastly, implementing these findings into real-time decision-making frameworks could provide coaches with actionable insights during live games, allowing for strategy adjustments based on in-game data, which could offer a competitive advantage.

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